Monotonic functions of a Random Variable (continued)

If X is a continuous RV, and g is an increasing function, then $Y = g(X)$ is a continuous RV and

$$F_Y(y) = F_X(g^{-1}(y))$$

Note: if g is decreasing, then, in general,

$$F_Y(y) \neq F_X(g^{-1}(y))$$

Example. Suppose X is a continuous random variable. Write the CDF ~~and PDF~~ of

$$Y = aX + b$$

in terms of the CDF ~~and PDF~~ of X. $g(Y) = aX+b$

(1) $a > 0$, then $aX+b$ is increasing in X.

$$F_Y(y) = P(Y \le y) = P(aX+b \le y)$$
$$= P\left(X \le \frac{y-b}{a}\right)$$
$$= F_X\left(\frac{y-b}{a}\right) \quad \leftarrow g^{-1}(y)$$

(2) $a < 0$, then $aX+b$ is decreasing in X.

$$F_Y(y) = P(Y \le y) = P(aX+b \le y)$$
$$= P\left(X \ge \frac{y-b}{a}\right)$$
$$= 1 - P\left(X \le \frac{y-b}{a}\right)$$
$$= 1 - F_X\left(\frac{y-b}{a}\right)$$
$$= 1 - F_X(g^{-1}(y))$$

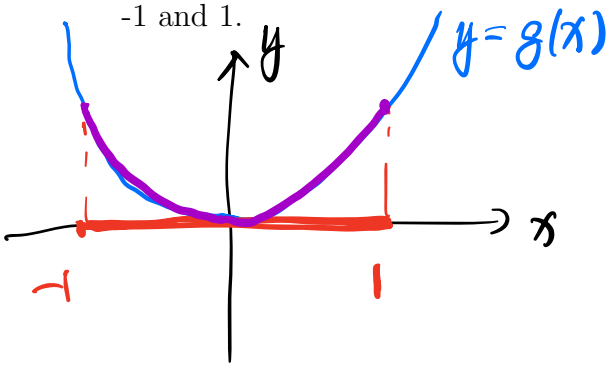
(3) $a = 0$, then $Y = b$.

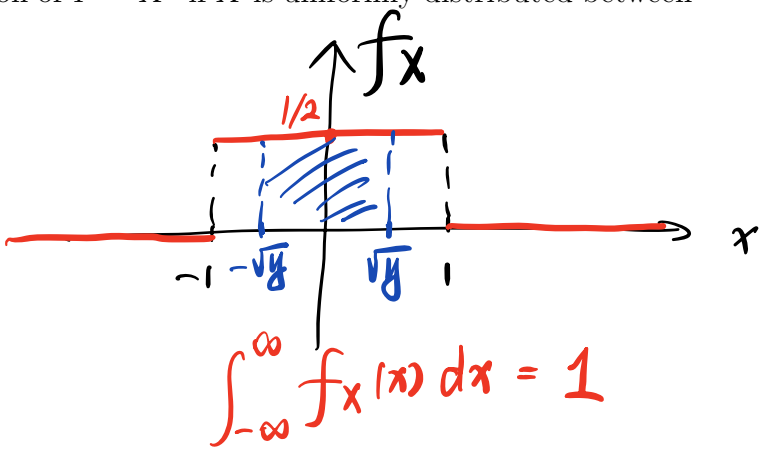
$$F_Y(y) = P(Y \le y) = P(b \le y) = \begin{cases} 1, & b \le y \\ 0, & b > y \end{cases}$$

Distributions of Non-Monotonic functions of a Random Variable

When a random variable X is transformed by a non-monotonic function g, the distribution of $Y = g(X)$ can be determined using the Distribution-Accumulation-Differentiation (DAD) method, which first identifies the possible values of Y, then determines its cumulative distribution function (CDF), and finally obtains its probability density function (PDF) by differentiation.

Example. Find the probability density function of $Y = X^2$ if X is uniformly distributed between -1 and 1.





$R_X = [-1, 1]$ (range of X)

Then $R_Y = [0, 1]$ (range of Y)

CDF of Y: $F_Y(y) = P(Y \le y)$

Case 1 $y < 0$

$F_Y(y) = P(Y \le y) = 0$

impossible event, since $Y \ge 0$.

Case 2 $y \in [0, 1]$

$$F_Y(y) = P(Y \le y)$$
$$= P(X^2 \le y)$$
$$= P(-\sqrt{y} \le X \le \sqrt{y})$$
$$= \int_{-\sqrt{y}}^{\sqrt{y}} f_X(x)\,dx = \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{2}\,dx = \sqrt{y}$$

Case 3 $y > 1$

since $Y \le 1$

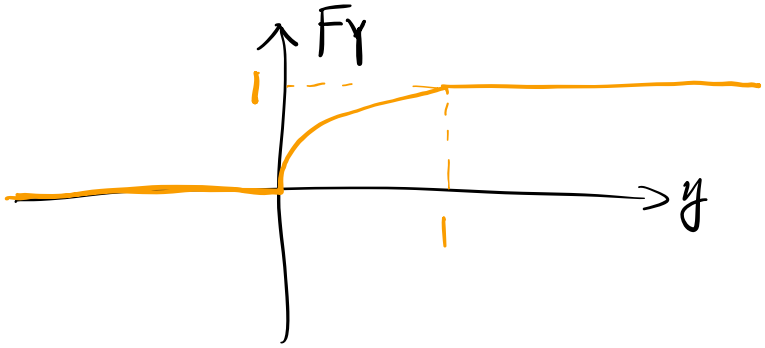
$$F_Y(y) = P(Y \le y) = 1$$

$$\left(= P(X^2 \le y)\right)$$

$R_Y = [0,1]$

Therefore, the CDF of Y is

$$F_Y(y) = \begin{cases} 0 & , \ y < 0 \\ \sqrt{y} & , \ 0 \le y \le 1 \\ 1 & , \ y > 1 \end{cases}$$



The PDF of Y:

$$f_Y(y) = \frac{d F_Y(y)}{dy} = \begin{cases} \frac{1}{2\sqrt{y}} & , \ 0 < y < 1 \\ 0 & , \ \text{otherwise} \end{cases}$$

The DAD Method

The Distribution-Accumulation-Differentiation (DAD) method provides a systematic way to determine the distribution of $Y = g(X)$, particularly when g is non-monotonic.

1. Distribution: Determine the range of $Y = g(X)$.
2. Accumulation: Find the cumulative distribution function (CDF) of Y:

$$F_Y(y) = P(Y \le y) = P(g(X) \le y).$$

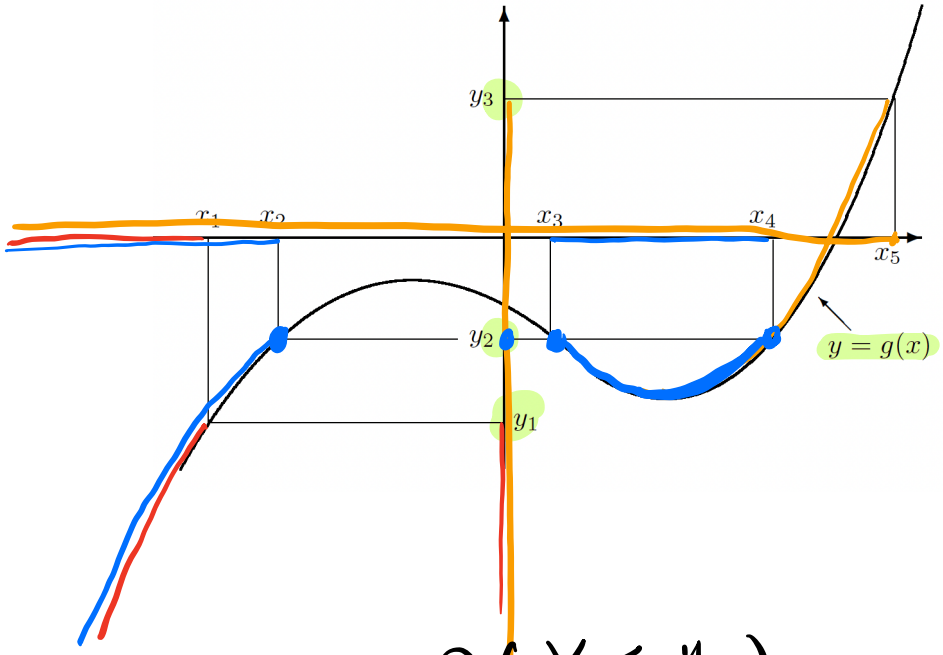
   Express this probability in terms of the distribution of X.

3. Differentiation: Differentiate the CDF to obtain the probability density function (PDF), where the derivative exists:

$$f_Y(y) = \frac{d}{dy} F_Y(y).$$

The DAD method does not require g to be monotonic or invertible. It can be applied to both monotonic and non-monotonic transformations of continuous random variables.

Example. Consider $Y = g(X)$ shown in the following figure. Determine $F_Y(y_1)$, $F_Y(y_2)$, and $F_Y(y_3)$ in terms of the distribution of X.



(1) $F_Y(y_1) = P(Y \le y_1)$

$= P(g(X) \le y_1)$

$= P(X \le x_1)$

$= F_X(x_1)$

(3) $F_Y(y_3) = P(Y \le y_3)$

$= P(g(X) \le y_3)$

$= P(X \le x_5)$

$= F_X(x_5)$

(2) $F_Y(y_2) = P(Y \le y_2)$

$= P(g(X) \le y_2) = P(X \le x_2) + P(x_3 \le X \le x_4)$

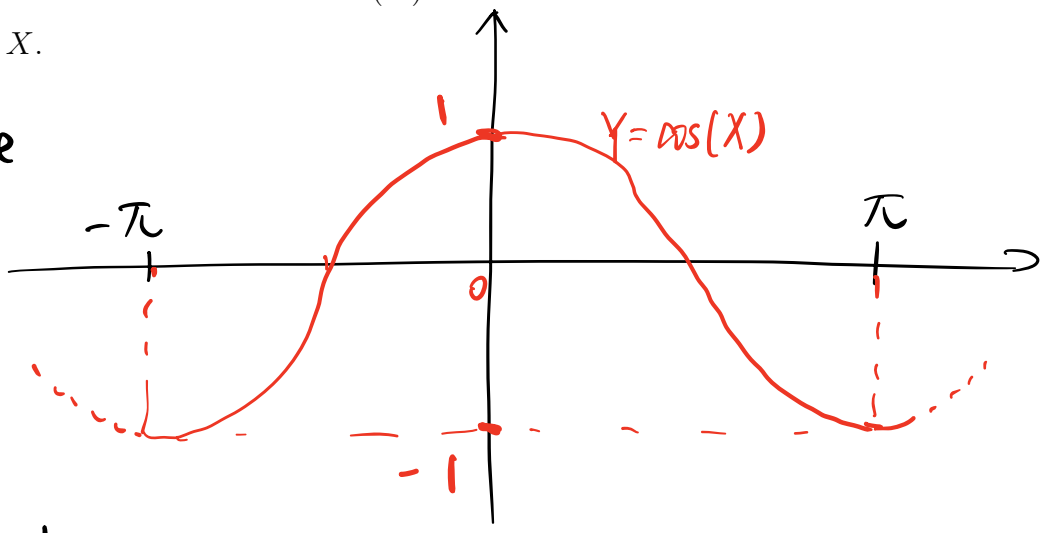
$= F_X(x_2) + F_X(x_4) - F_X(x_3)$

$g(X) \le y_2 \iff x_3 \le X \le x_4$ or $X \le x_2$

Example. Suppose X is a continuous random variable taking values in $[-\pi, \pi]$. Write the CDF of

$$Y = \cos(X)$$

in terms of the CDF of X.



Determine the range of Y:

since $-\pi \le X \le \pi$,

we have

$$-1 \le Y = \cos(X) \le 1$$

Case 1 : $y < -1$

$$F_Y(y) = P(Y \le y) = P(\cos(X) \le y) = 0$$

$-1 \le \cos(X) \le 1$ & $y < -1$

Case 2 : $y > 1$

$$F_Y(y) = P(Y \le y) = P(\cos(X) \le y) = 1$$

$-1 \le \cos(X) \le 1$ & $y > 1$

Case 3 : $-1 \le y \le 1$ (to be continued ……)